

Problem Statement Selection

Problem statement addressed: Cleveland Clinic — *Unlocking undruggable targets: quantum simulation of allosteric signal propagation* (Global Quantum + AI Challenge 2026, Cleveland-Clinic-Challenge-Statement-vF-1.pdf).

a reader holding only the published version would otherwise see unexplained deviations from Table 1. Each reply, and what it changed here:

These replies granted latitude on **structure choice and deliverable format**. They did not endorse any finding in this submission, and no finding here is presented as endorsed.

How we interpreted it

Rather than only running the requested algorithm on the provided structures, our Phase-1 work audited the challenge’s own premise and dataset. Our measurements show that the standard benchmark targets conflate **static cavity retrieval** with **genuine cryptic-pocket discovery**: roughly half of apo/holo pairs cannot express the required contrast at all.

Our proposal therefore addresses the challenge’s core objective by delivering a **validated, unconfounded benchmarking instrument** capable of certifying allosteric methods — quantum or classical — alongside the required predictions.

Targets

We address the complete mandatory set required by Challenge Statement §6: **KRAS G12C, BCR-ABL1, cardiac myosin, and c-Myc**.

Three structures deviate from Table 1 (4LDJ replacing 4OBE; 8QYP/8QYR replacing 5TBY/6C1H); BCR-ABL1’s 10PL is retained, not substituted. **Each is stated, with its reason and its evidence, in Section 1 of the Concept Proposal**, and each traces to the organisers’ clarification of 2026-08-26. They are not repeated here: a single authoritative statement is less likely to drift than two, and the Concept Proposal is the document a reviewer assesses.

Required outputs

Challenge Statement §5’s three deliverables are supplied as follows:

Note on organiser correspondence

Included at the organisers’ request, so that this submission is neither advantaged nor disadvantaged relative to teams working from the published Challenge Statement alone.

Several choices below follow from replies the Cleveland Clinic team sent in answer to questions we raised. The Challenge Statement was not publicly revised, so

§ 5 output	Where
Connectivity Matrix — N×N quantum connectivity strength	Connectivity_Matrices.csv
Hit List — top five predicted allosteric residues per target	Solution_Outputs.pdf , and machine-readable in the same file set
Methodological Report — the quantum metric chosen, and why it proxies biological signal transmission	Section 2 of the Concept Proposal , with per-target detail in Solution_Outputs.pdf

organiser reply	what it changed in this submission
2026-08-26 — cardiac myosin: our 8QYP–8QYR substitution is <i>accepted as primary</i>	Cardiac myosin is scored on 8QYP/8QYR rather than Table 1’s 5TBY/6C1H
2026-08-26 — a suggested KRAS G12C structure, 8S8C	We checked 8S8C against the PDB, found it holo (MK-1084-bound) and therefore unusable as the apo half of a contrast, and substituted verified apo 4LDJ instead
2026-08-26 — BCR-ABL1: 10PL may be substituted with an alternative apo structure, <i>“please document the rationale in your submission”</i>	We declined the substitution and retained 10PL. Rationale, as requested: its myristate occupancy is the finding, not a defect — it is how we established that “apo” depositions are not reliably ligand-free. Stated in Section 1 of the Concept Proposal
2026-08-26 — no specific deliverable formats are prescribed; use formats accessible with conventional software	The five-file package structure, and the connectivity matrices shipped as one long-format CSV rather than an archive
2026-09-07 — submitted documents may be revised and re-uploaded	Nothing in the science; recorded for completeness
2026-09-08 — ENM methods are permitted, as is an MD- <i>trained</i> tool used with MD-free inference	PocketMiner retained as the cryptic-opening veto